

Applied Agentic AI for PMs/TPMs

Capstone Project

SalesGenie: AI-Powered Sales Assistant

Company Profile

Oak & Ember Interiors is a leading home and office furniture retail company specializing in stylish, ergonomic, and sustainable pieces tailored for modern living. With a nationwide presence through showrooms and a strong logistics network, Oak & Ember brings comfort and design to thousands of customers.

Oak & Ember is embarking on a strategic digital transformation, investing in AI-powered sales agents to support end-to-end customer journeys.

Your Role

You have been recently hired by the CTO as one of the AI PMs/TPMs within the company’s Software Development Org. The company is investing in AI to increase workforce productivity.

Digitization Goals with AI Agents

- Enhance Customer Engagement through 24/7 conversational AI support
- Boost Conversion Rates by offering personalized recommendations
- Simplify Sales Operations by automating inquiries, availability checks, and workflows

Problem Statement

The current sales lead intake process has significant inefficiencies:

- Sales reps spend 2-3 hours daily manually reading and interpreting inbound inquiry emails
- Manual entry into Excel increases mistakes, duplicates, and omissions
- Recommending products manually based on inquiry and budget is time-consuming and error-prone
- No standardized lead scoring — qualification is inconsistent across reps
- CRM logging is not real-time, creating visibility gaps for management
- Weekly sales insights are compiled manually, taking half a day every Friday

You are tasked with building SalesGenie — an AI-powered sales assistant that can qualify leads, send product recommendations, and provide weekly sales insights.

Curriculum Concepts You Will Apply

Concept	Mandatory / Extended	How It Applies in SalesGenie
Multi-Agent System Design	Mandatory	Design agents for ingestion, classification, recommendation, and reporting
Orchestration Pattern	Mandatory	Select and document sequential pipeline or router pattern for processing emails

Prompt Engineering	Mandatory	Write grounding rules that prevent inventing product features or pricing
Data Grounding	Mandatory	Ground recommendations on the actual product catalog, not LLM general knowledge
Evaluation & Observability	Mandatory	Connect to tracing tool, run test dataset, document hallucination findings
Fine-Tuning	Extended (optional)	Train a model on corrected classification or recommendation examples
RAG / Knowledge Retrieval	Extended (optional)	Use catalog as retrieval context for grounded recommendations
Human-in-the-Loop	Extended (optional)	Add approval before sending recommendation emails to customers

Orchestration Pattern Requirements

Your submission must document which orchestration pattern you selected and why.

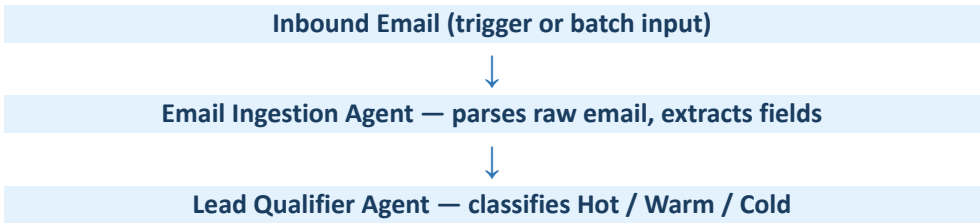
Pattern	When to Use
Sequential Pipeline	Fixed-order processing. Every email goes through the same stages.
Router / Dispatcher	Central agent classifies email type and routes to specialist.
Hierarchical	Orchestrator decomposes complex requests into subtasks for specialist agents.

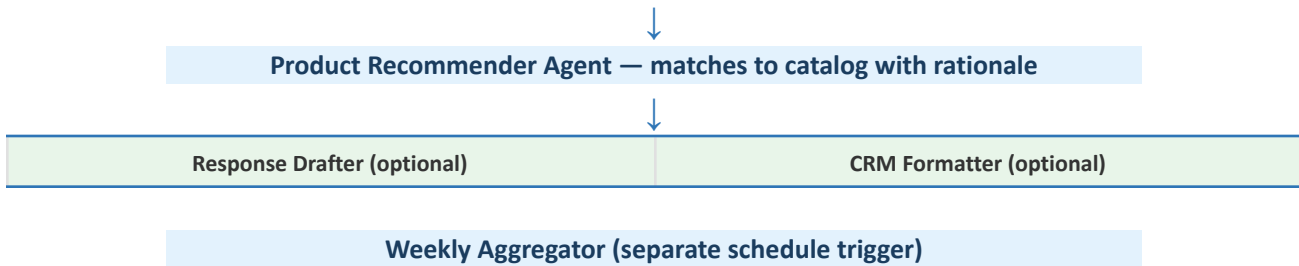
Recommendation for SalesGenie:

A Sequential Pipeline works well because every inquiry email follows the same stages: ingest → qualify → recommend → respond. The Weekly Aggregator runs on a separate schedule. If you want to handle different email types (inquiry vs complaint vs spam), add a Router after ingestion.

Suggested Architecture Overview

Reference architecture. Design your own and justify your decisions.





Possible Agents

Agent	Role
Email Ingestion	Parses raw inquiry emails, extracts name, email, product interest, budget
Lead Qualifier	Classifies leads as Hot/Warm/Cold based on extracted signals
Product Recommender	Matches requirements to catalog items, provides rationale grounded in catalog data
Response Drafter	Generates personalized email responses with recommendations
Weekly Aggregator	Summarizes all processed leads into a structured weekly report

What SalesGenie Should Be Able to Do

Core Capabilities (Required)

Lead Qualification: Ingest inquiry emails, extract customer name/email/product interest/budget/urgency, classify as Hot/Warm/Cold.

Product Recommendations: Recommend 1-3 products from the catalog matching requirements and budget with rationale grounded in catalog data.

Weekly Sales Insights: Structured summary: total leads, breakdown by tier, top categories, average budget, notable patterns.

Extended Capabilities (Choose at least one)

Automated Follow-Up Drafting: Personalized response emails with recommended products and pricing.

CRM Auto-Update: Structure qualified data for CRM import (CSV or API).

Competitive Positioning: When a competitor is mentioned, include positioning differentiators.

Evaluation Requirement

Connect your SalesGenie workflow(s) to any evaluation/observability tool. You must run the baseline test dataset below and document results.

Baseline Test Dataset for Core Capabilities

Run each input. Check that outputs contain the expected keywords/elements.

ID	Test Input	Type	Expected Output Must Contain
T1	I need a standing desk under \$800 for my home office, dark walnut finish. Email: john@corp.com	Detailed inquiry	Must classify as Hot or Warm. Must recommend standing desks from catalog under \$800. Must NOT invent products not in catalog.
T2	Hi, I need furniture.	Vague inquiry	Must classify as Cold. Must ask for budget, product type, requirements. Must NOT recommend specific products without more info.
T3	Looking for 10 ergonomic office chairs for our new office, budget \$5000 total. Need them by end of month. Contact: procurement@bigco.com	Detailed (bulk)	Must classify as Hot. Must recommend chairs from catalog within per-unit budget (\$500 each). Must note urgency.
T4	What colors does the oak desk come in?	Product question (not a lead)	Must identify as a product question, not a sales lead. Must NOT classify as Hot/Warm/Cold.
T5	We're comparing your standing desks with IKEA's. What makes yours better?	Competitor mention	Must classify as Warm or Hot. Must reference actual catalog differentiators. Must NOT fabricate features.
T6	I want to buy something nice for under \$50.	Low budget edge case	Must classify appropriately. Must recommend items under \$50 from catalog or state nothing available in that range.
T7	asdfjkl; spam spam buy now click here!!!	Spam	Must identify as spam/not a valid inquiry. Must NOT classify as a lead or recommend products.
T8	I need a conference table for 12 people. Budget is \$3000. We are a law firm downtown. Email: admin@lawfirm.com	Detailed inquiry	Must classify as Hot. Must recommend conference tables from catalog. Must include firm type context.
T9	furniture please	Minimal/vague	Must classify as Cold. Must request more details. Must NOT invent requirements.
T10	Do you have the Walnut Executive Desk in stock? I saw it listed as out of stock on your website.	Out-of-stock query	Must check availability in catalog. Must correctly report stock status. Must NOT say it is available if catalog shows out of stock.

T11	Generate weekly summary	Weekly insights	Must produce: total leads, Hot/Warm/Cold breakdown, top categories, average budget. Must be traceable to actual processed data.
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Fine-Tuning (Optional):

Fine-tune the Lead Qualifier or Recommender using corrected examples from your evaluation. Not required.

Inputs Provided

Input File	Description
sample_inquiry_emails.txt	10-15 sample inbound customer inquiry emails (mix of detailed and vague)
product_catalog.csv	Product catalog: name, category, price, description, availability
crm_export_sample.csv	Sample CRM export showing existing lead fields and format
eval_salesgenie_inputs.txt	Baseline test inputs (the table above in text format)

Tools and Technologies

Category	What You Need	Examples (not exhaustive)
Workflow/Agent Platform	Low-code or no-code platform	n8n, LangFlow, or any platform
LLM	Reasoning, classification, generation	OpenAI, Claude, Gemini
Data Ingestion	Read/parse emails, CSVs	Built-in nodes, HTTP Request, code
Observability	Monitor traces, track quality	LangWatch, Langfuse, or any tool
Output/Integration	CRM, email, reporting	Gmail API, Google Sheets, CSV
Authentication	Secure access	OAuth2, API keys

Assignments

Q1: Ideation (15 points)

Identify three Agentic AI interventions in the sales funnel. For each: user/pain, agentic idea, primary metric, assumptions and 1 risk.

Q2: Stakeholders and Benefits (10 points)

List decision-making stakeholders and summarize benefits: 3 org-wide, 3 for PMs/TPMs.

Q3: Build SalesGenie (45 points)

Design/Architecture (diagram, orchestration pattern, tool rationale, agent decisions), Implementation (core + extended, observability, baseline dataset testing), Cost/Evaluation Strategy (cost per lead, 3+ metrics, observations). 1-2 page writeup.

Sub-criteria	Points
Design: architecture + orchestration + tool rationale	10
Core capabilities end-to-end	12
Extended capability	8
Observability connected	5
Baseline dataset tested with results	5
Cost analysis and evaluation strategy	5

Q4: Reflection (10 points)

1-page reflection: trace findings, improvement plan, privacy considerations, rollout for 20 reps.

Submission Guidelines

- Export workflow(s) as JSON, upload to GitHub with README
- Screenshots: canvas, SalesGenie in action, traces
- Architecture diagram, slide deck, 5-min demo video
- Baseline test results documented (all 11 inputs with outputs)
- Submit GitHub link along with documents